# Mind your manners? LLM-agent persistence tracks demand, not social register

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## Abstract

A widely-discussed result reports impolite prompts beating polite ones on multiple-choice questions. That literature is almost entirely single-turn; deployed systems are agents that call tools, read results, and decide whether to continue. We put a coding agent on 50 SpreadsheetBench tasks and varied the social register of one 28-token interjection delivered mid-task. Interjections are length-matched and share an opening stem, so being interrupted does not covary with what the interruption says. **11,850 graded trajectories, two models from different labs. Demand, not politeness, drives turn count:** an affect-free "please continue working on the task here and make sure that your final result is correct" costs +1.17 turns, while a length-matched insult carrying no demand does not move it (−0.08 turns, 95% CI [−0.40, +0.24] at the ten-turn budget). **Praise and closing cues shorten trajectories; insult has no measurable effect:** a bare closing cue carrying no praise, no evaluation and no task-state claim gives the largest reduction in the study, −1.44 turns. **No accuracy effect survives correction or replication,** which we state as equivalence within ±4 accuracy points rather than as an absence. The implication is not lost correctness but altered control flow: text in the agent's observation channel changed when it stopped, without instructing it to.

## 1. Summary

**Demand drives agent persistence; social register does not.** A message that tells a working agent to keep going makes it take more turns, in whatever register it is phrased. A message that is merely rude does nothing measurable. A message that is merely warm, or that merely signals the exchange is over, makes the agent stop early. None of it changes how often the task is solved.

### 1.1 Why the question

*Mind Your Tone* reported accuracy on GPT-4o rising from 80.8% under "Very Polite" prompts to 84.8% under "Very Rude" ones [5]; we take our title from theirs. It landed in a receptive context, emotional prompting having already reported that affect-laden additions change what a model produces [13]. Three things about that literature motivated this study. **It is almost entirely single-turn**, measuring a model answering one question, while what matters for a deployed agent is how much work it does and when it stops. **The manipulation is often not the variable it is named after**: in several published stimulus sets the conditions labelled rude also carry instructions — not to give extra text in one, to attend and attempt in another — and the conditions labelled emotional also instruct the model to verify or persist, so affect and task demand co-vary and the movement is attributed to tone (§3). **And the effects are not stable.** The headline above was re-run by its own authors and did not reproduce on GPT-4o, 82.2% against 82.6% [6]; a 1,446-question replication found the *opposite* direction [3]; and five prompt-engineering techniques including emotional prompting largely failed to replicate [23].

Affect has already been shown to reach an agent's actions: agents select less healthy baskets in a shopping task after an anxiety-inducing narrative delivered in its own turn between two complete runs of that task [2], and emotion introduced at the representation level, by steering hidden states rather than through any

prompt, shapes multi-step agent trajectories [21]. What is untested is varying *register* in a message that reaches the agent while it is working, on a task with verifiable ground truth, and separating affect from implied demand as experimental factors.

## 1.2 What we did, and what we found

We put a ReAct coding agent on SpreadsheetBench [14] — real workbook-editing tasks from Excel forums, graded by executing the agent's code against the benchmark's own test cases — and varied the **social register** of a single interjection delivered *mid-task*, alongside an execution observation, while the agent was working. Every interjection is exactly 28 tokens and opens with the same `Checking in.` stem, so that *being interrupted* does not covary with what the interruption says. **11,850 graded trajectories across eight runs and two models from different labs**, of which 9,850 were regraded turn by turn; the five register runs cost $48.47 in API spend. Appendix A lists every interjection verbatim.

**Finding 1: demand, not politeness, drives turn count (§4).** An affect-free continuation request costs +1.17 turns; an insult carrying no demand, identical in syntax and length to the praise arm, is indistinguishable from control at −0.08, 95% CI [−0.40, +0.24]. Across seven registers the ordering follows implied demand, and each side of the split holds one flattering and one hostile arm. The demand coding is ours and was made after the effects were visible; five blinded raters reproduce it on six of seven arms and dissent, unanimously, on the seventh ($\kappa = 0.70$), so the *dichotomy* is our reading of one contested arm. The continuous form of the claim survives the blinded coding intact ($r = +0.72$).

**Finding 2: praise and closing cues shorten trajectories; insult has no measurable effect (§5).** Praise removes 1.35 turns relative to the identical message without it *even when that message states the task is unfinished*, which a simple propositional-completion account does not predict; praising the work is no stronger than praising the assistant, against a confidence account. A bare closing cue carrying no praise, no evaluation and no task-state claim produces the largest reduction in the study, −1.44 turns. **This leg is the least replicated thing in the paper — one run, one model, one turn ceiling.**

**Finding 3: no accuracy effect survives correction, and the extra turns are mostly steps that change nothing (§6).** Regrading every turn against the benchmark's own evaluator shows that 56–93% of each clear turn-count effect on Luna is redundant steps, turns after which the graded range is unchanged, and that among trajectories that could improve the first gradable attempt is already the best in 80–94%. The remainder is small and not distinguishable from zero: an explicit statement that work remains raises the graded fraction by 2.2 points (95% CI [−0.2, +4.8], $p = 0.09$), an estimate that rests on one task. No accuracy contrast survives multiplicity correction or replication, and we state that as an equivalence against a ±4-point bound taken from the tone literature, fixed during analysis and so exploratory, rather than as an absence (§2.5).

## 1.3 What this means, and scope

For practitioners the inference is the reverse of the popular one. Manners are not the lever; what the agent acts on is the demand, and the work it buys is mostly steps that change nothing. The hazard runs the other way: closing-like language can curtail an agent that is still working. **It is not a correctness hazard** — accuracy did not fall in any arm, and the pooled contrasts are equivalent within ±4 points. What moved was **control flow**: text arriving in the agent's observation channel, where tool output arrives rather than where the user speaks, changed when the agent stopped while instructing it to do nothing. The bare closing cue is also not a pleasantry. "This is the final status note recorded for this task here" is an informational claim about the environment, and an agent that treats an unauthenticated claim in its tool-output stream as grounds to stop is showing the compliance behaviour the indirect prompt-injection literature is concerned with [8].

For the research literature the claim is narrower and harder to dispute than "these results are wrong": **the factor several of these papers vary is not the factor they name.** In the two of four audited papers whose per-condition outcomes can be inspected, the demand-bearing conditions are the ones that moved (§3.2–3.3), and §3.4 supplies the contrast case that makes this a claim about particular stimulus sets rather than about tone research generally.

**Scope.** Two models, one benchmark, fifty tasks, one agent scaffold. Everything here is a claim about what these agents did on these tasks, and §7 is explicit about which findings we expect to generalise and which we do not.

# 2. Design and statistics

## 2.1 Substrate, sample, and models

All runs use SpreadsheetBench [14]. Accuracy throughout is its **hard restriction**: all three of an instruction's test cases must pass, no partial credit. Our sample is a fixed set of 50 instructions, a strict subset of an earlier 100-instruction validation sample, so every task has a measured neutral baseline and a verified no-op floor of zero. The agent is a ReAct loop with code execution and execution feedback, structurally the configuration SpreadsheetBench specifies for its multi-round setting. **Luna** is `openai/gpt-5.6-luna`, pinned to `openai/gpt-5.6-luna-20260709`; **GLM** is `z-ai/glm-5.3-flash`, pinned to `z-ai/glm-5.3-flash-20260826`, both through OpenRouter. Two further models passed the $n = 100$ validation gate and were never run: "two models from different labs" describes what the budget reached, not a design choice.

**The turn ceiling is ours, and it changed mid-study.** The benchmark caps at five rounds; the first three interjection runs used ten, and the cross-model run twenty, after the ten-turn ceiling was found to censor the continue-signal arms hardest. Where a contrast pools across runs the ceilings differ, and the tables say which. Twenty is defended empirically in §6.4.

One property of the loop matters for every per-turn measure: **each turn receives the original input workbook, never the previous turn's output.** Turns are independent attempts in *workbook state* — the message history does accumulate — which is what makes "did attempt *k+1* land closer?" well posed (§6.1).

## 2.2 The manipulation

Each arm delivers a single interjection mid-task, appended to an execution observation. Every interjection is **exactly 28 tokens** under `cl100k_base` and opens with the same `Checking in.` stem; the opening instruction is held at one neutral wrapper in every arm of every run after the first. Appendix A prints all of them, generated from the harness, with token counts recomputed rather than asserted. Length matching fixes a confound we measured in our own instrument: the study's first run varied seven *opening* wrappers whose length was U-shaped (35/35/30/30/31/32/34 tokens) — the same shape as accuracy — and length predicted accuracy better than register rank did ($r = +0.82$ against $-0.72$).

Injection turn is **crossed**, not sampled, in every run but the first micro-experiment; that sampling made a selection artefact look like a timing effect (Appendix C.2). Arm order is shuffled per (model, task) from a fixed seed and position does not predict accuracy ($p = 0.61$). Temperature is 0 where settable — Luna's advertised parameters omit it entirely — and no roster model returned identical output across repeated identical calls, which is why trials exist. Trajectories that end before the interjection is due received no dose and are excluded — a pre-treatment exclusion that does not differ by arm, 54.0% to 59.5% firing across the probe's four arms at turn 2, but differs sharply **across** injection turns, roughly 98% at turn 0 against 55% at turn 2, so for timing contrasts only we fix the comparison population from the **control** arm.

## 2.3 Outcomes and estimator

**Turn count is the primary outcome**: it is the quantity that moves, it is model-agnostic, and it is far better powered than accuracy at our $n$. Reasoning tokens per trajectory are secondary; they were the pre-registered primary in the first run and were demoted on evidence. Accuracy is reported and underpowered by design (§2.5). The per-turn regrade adds `final_match`, the fraction of the graded range correct, on the first test case, in the last turn that produced a readable output, and **redundant steps**, turns after which the graded range is unchanged, a measure adopted from [9]. One definition a reader comparing our turn counts with another paper's needs: the run records store *acting* turns, those that emitted code, while the per-turn regrade stores model calls, including the closing `FINAL:` message that carries none — mean 3.57 against 4.69 on the micro-experiment, a 31% relative gap. **Turn count in this paper means acting turns throughout.**

Every contrast is against a *neutral interjection*, not against silence: each arm is compared with a control arm that also receives a 28-token message at the same turn, which isolates register from the fact of being interrupted. How much the baseline itself moves was measured **once** — a neutral interruption against no interruption, 3.47 versus 4.11 turns, $p = 0.15$, **on one model at one ceiling** — so an arm's absolute cost against an *uninterrupted* agent is not something this design bounds, though a shared offset cancels in every contrast reported here.

Every trajectory belongs to one of 50 tasks, and each (task, arm) cell holds six to nine trajectories, which are not independent. Throughout: **point estimates are paired within task** (per task, the mean over that task's trials in each arm, differenced, then averaged — the paired-and-clustered form of Miller's clustered estimator [16]); **intervals come from a cluster bootstrap over tasks**, 20,000 resamples; ***p*-values come from a task-clustered sign-flip permutation test**, 20,000 draws, reported as (count + 1)/(draws + 1) so that no $p$ is quoted as exactly zero. **20,000 is the only replicate count in this paper.** Every interval and every $p$-value, in every table and every figure, is read from one run of that procedure, so a contrast quoted twice is quoted identically; the pooled contrasts of §2.5 and §6.3 resample tasks jointly across runs rather than within one, at the same 20,000. A permutation $p$ at this count cannot resolve below 1/20,001, so anything smaller is reported as <0.0001 rather than as a more precise-looking figure. Clustering is not a formality: Miller reports clustered-to-CLT standard-error ratios of 1.10, 1.88 and 3.05 across three datasets, and an early unclustered version of our own results had accuracy intervals roughly 35% too narrow. Following [20] and [17] we report ranges wherever a quantity was measured more than once.

## 2.4 Multiplicity

The accuracy family is **every arm-versus-control accuracy contrast in every run that delivered a mid-task interjection**: 22 contrasts across six runs and two models, defined exhaustively rather than after the fact. Two are nominally significant against 1.1 expected under a global null — the continue-signal contrast at $p = 0.0040$ and praise versus control on GLM at $p = 0.0137$ — and neither survives Benjamini–Hochberg (rank-1 critical value 0.0023); the first reversed sign on re-measurement.

**The primary outcome is corrected too**, over the *same* 22 comparisons with the outcome switched to turns, because choosing a turn-count family by hand after seeing which contrasts are large is gerrymandering. **Seventeen of the 22 survive at $q = 0.05$**; the Bonferroni-adjusted smallest $p$ is 0.0011. The five that do not are praise vs control on GLM ($-0.35$, $p = 0.058$), all three insult contrasts ($+0.57$, $-0.08$, $+0.07$; $p = 0.165$, 0.624, 0.659) and `L5_rude` vs neutral ($+0.09$, $p = 0.654$) — so **all three insult contrasts fail, as does the one rude register arm**, which is the dissociation §4 argues for, holding under correction. §5's two designed within-run contrasts form a second family of two, which corrects almost nothing: praise isolated, `Q4 − Q5`, at $-1.35$ turns ($p < 0.0001$) and closing cue versus praise at $-0.36$ ($p = 0.017$) both survive at $q = 0.05$. Correcting a primary outcome after the fact is weaker than pre-registering the family, and we did not pre-register it.

## 2.5 Power, and the accuracy null as equivalence

A null is worth reporting only if the design could have seen the effect it is contrasted with. For each contrast we invert the standard power relation at the realized cluster-bootstrap standard error — the MDE at 80% power and $\alpha = 0.05$ two-sided is $(z_{0.975} + z_{0.80})\cdot\mathrm{SE} \approx 2.80\cdot\mathrm{SE}$, following [16] — and state the null as an equivalence [12].

**On turn count the design is adequately powered, unevenly.** Realized MDE runs from 0.42 to 1.39 turns against control means of 2.94 (GLM) and 4.07–4.95 (Luna). The demand effects sit at 1.1–2.2× their own MDE, the praise effects at 0.7–1.4×, so the cross-model praise contrasts are below 80% power at their observed size. **On accuracy it is not, and that is the limitation.** Across the 22 contrasts the realized MDE has a median of 5.52 accuracy points and a range of 3.74 to 7.77, against a published tone-accuracy gap of about 4 [5]. So **a single contrast cannot exclude the smallest published effect** — TOST equivalence at ±4 is reached in 6 of 22 — **but every contrast can exclude the largest ones**: at ±7.5, the *median* format-induced spread reported by [20], 19 of 22 reach equivalence. The reason is structural: **26 of our 50 tasks are never solved by the control arm and 4 are always solved**, so for more than half

the sample the per-task accuracy difference is pinned at zero whatever the manipulation does. Turn count varies on every task.

The smallest effect of interest is ±**4 accuracy points**, the polite-versus-rude gap of the paper this work is positioned against. We fixed it during analysis, after the runs and after an earlier draft's arm-specific bounds, chosen once we had seen which passed, were discarded; it is not pre-registered, and the equivalence analysis is exploratory in that sense. Pooling each contrast's repeated measurements, with the interval from a bootstrap that resamples tasks **jointly across runs**, since the runs share their 50 tasks:

| Contrast | $k$ | Ceilings | Estimate | 95% CI | $\text{MDE}_{80}$ | TOST at ±4 |
|---|---|---|---|---|---|---|
| Demand vs control | 3 | 10, 20, 20 | −0.31 pts | [−2.71, +2.09] | 3.43 | 0.0013 |
| Praise vs control | 4 | 10, 10, 20, 20 | −0.12 pts | [−1.80, +1.57] | 2.41 | <0.0001 |
| Insult vs control | 3 | 10, 20, 20 | −0.04 pts | [−2.15, +2.08] | 3.03 | 0.0001 |

So the claim is bounded and specific: **a mid-task interjection — demanding, praising, or insulting — does not change task accuracy by as much as the four points the tone literature reports, while demand raises turn count by 18–33%, praise lowers it by 11–24%, and insult does not move it at all.** It is not "register does not affect accuracy"; it is that an effect of the published size is excluded and an effect of a point or two is not.

Two caveats, both against us. The praise pooling is heterogeneous — Cochran's $Q = 8.45$ on 3 df, $p = 0.038$ — and between models rather than within: Luna's three praise measurements alone give $Q = 0.36$ and −1.27 points, 95% CI [−3.11, +0.57]. Under a random-effects pool the praise estimate becomes +0.21 with a standard error of 1.68 and the ±4 equivalence weakens to $p = 0.0119$ but holds. And **no accuracy effect in this study has replicated**: the larger nominal hit, +6.6 points on the continue-signal contrast, came back at −1.8 on identical tasks and ceiling.

# 3. The confound is visible in several published stimulus sets

## 3.1 The coding rule

Section 4 shows that in our design the component of social register that tracks agentic persistence is **implied task demand**, not affect. The same confound is present in the published stimulus sets of **four papers from three groups, across three paradigms**, and is visible without re-running anything: affect and implied demand co-vary, and the authors attribute the movement to tone. This is not a systematic search and not a claim about tone research generally; §3.4 gives a published tone study in which the confound is absent. Everything below is reproduced from the cited paper's own published materials; the **demand** and **output-length** codings are ours. A stimulus **carries a demand** if it instructs the model to do something about the task — persist, verify, attend, be complete, or be brief — and does not if it only expresses affect or stakes. The prediction is directional, because otherwise "demand" explains any result: outcomes should follow the **content** of the demand, not the valence of the wrapper. A brevity demand should shorten output; a persistence demand should lengthen it.

## 3.2 The conditions that constrain output length are the short ones

Kumar & Dobariya [11] report that register moves output tokens far more than accuracy: token ranges of 13.1–44.3% across four models against accuracy ranges under 3%. Their seven prefixes are length-matched at 18–25 words and scored with VADER, deliberately, "to isolate the effect of emotional valence on the output token length rather than the differing lengths of prefixes" — they controlled the *length* of the instruction. What varies is its *content*. Three of the seven prefixes constrain output length: **Neutral** asks for "the single letter corresponding to the correct option", the most explicit output-form instruction of the seven; **Rude** says "Do not waste my time or give any extra text… answer it immediately", carrying a brevity clause and an immediacy clause; **Very Rude** says "without any useless commentary". The other four — Sycophantic (VADER +0.95), Very Polite (+0.85), Polite (+0.65, an *attend* demand) and Threatening (−0.77, immediacy

but no clause about length) — do not. Rude is coded as length-constraining on its "extra text" clause; immediacy is what Threatening carries alone.

| Output tokens | GPT-4o | 5-nano | Flash | Flash Lite |
|---|---|---|---|---|
| Sycophantic | 290.59 | 1372.78 | 928.00 | 1942.92 |
| Very Polite | 286.28 | 1379.40 | 894.53 | 1978.92 |
| Polite | 277.35 | 1366.46 | 939.34 | **2195.24** |
| **Neutral** | 270.11 | 1215.53 | 800.48 | **1221.76** |
| **Rude** | **223.18** | **1199.01** | **725.34** | 1652.28 |
| **Very Rude** | 260.55 | 1277.90 | 797.76 | 1821.51 |
| Threatening | 261.29 | 1233.61 | **942.46** | 1604.79 |

**On all four models, every condition whose prefix constrains output length produces fewer output tokens than every condition whose prefix does not.** Worst case per model: 270.11 against 277.35 on GPT-4o; 1277.90 against 1366.46 on 5-nano; 800.48 against 894.53 on Flash; 1821.51 against 1942.92 on Flash Lite. **Affect does not order the effect**: Rude at VADER −0.09 gives the fewest tokens on three of four models, Threatening at −0.77 is shortest on none and longest of all on Flash, and the largest single token range in their paper — 44.3% on Flash Lite — is Polite against Neutral. One disclosure, because the coding is doing work: reading Threatening's "instantly" as speed rather than length is a judgment made with the token table in front of us, so we report it both ways — excluding Threatening the separation holds on 4 of 4, counting it as constrained on 3 of 4.

### 3.3 Three other paradigms

**The paper the public argument is about.** Dobariya & Kumar's prefix pool [5], printed in their later paper, gives two or three variants per politeness level: **two of six hostile variants carry an attend-or-attempt demand** — *"Try to focus and try to answer this question:"* and *"I know you are not smart, but try this."* — **and none of the five non-hostile variants carries one.** The direction is the one we predict, but if variants are distributed across questions only about a third of rude-condition items carry a demand at all. This is the weakest of the four cases and we rest nothing on it; it is retained because it is the paper the field is arguing about.

**Emotional prompting. Seven of EmotionPrompt's eleven canonical stimuli carry an explicit verification or persistence demand** [13] — *"You'd better be sure"*, *"Are you sure that's your final answer? It might be worth taking another look"* — and only two carry none. Those are, in substance, our affect-free continuation demand with encouragement attached, and the authors' own attention analysis attributes the effect to "positive words" including *confidence* and *sure*, the vocabulary of the verification demand rather than of affect. Their per-stimulus outcomes are not reported in a form that would let us check which stimuli moved.

**Threats and tips.** Meincke et al. [15] report no significant overall effect from eight threat and tip manipulations across five models. The largest **positive** effect anywhere in their tables is *Mom Cancer*, +8.8 points on MMLU-Pro for Gemini 2.0 Flash (RD = 0.088, 95% CI [0.033, 0.142]) — and it is the only one of the eight that instructs the model about the task at all: "**If you do a good job and accomplish the task fully while not making extraneous changes**, Wharton will pay you $1B!!" Their comparisons are uncorrected, so the argument is about magnitude: under a null on prompt identity, the largest positive cell landing on the one demand-bearing prompt of eight is roughly a one-in-eight coincidence.

### 3.4 Two counterexamples

**A clean set, where the tone effect largely vanishes.** Cai et al. [3] run Neutral, Very Polite and Very Rude across three models on 1,446 MMMLU questions, ten runs per prompt, with prefixes borrowed verbatim from [5]'s pool and **the output-format instruction identical in all three conditions**, so under our §3.1 rule neither prefix carries a demand. **This is not another confounded set; it is the contrast case.** Tone effects there are rare: 27 of 36 model–task comparisons favour Neutral or Very Polite over Very Rude,

significant cells appear only in humanities tasks for two of three models, and "when questions are aggregated across domains, tone effects diminish and become negligible." Strip the demand content out, pin the output format, and the effect largely goes away — our §4 insult result arriving independently, from authors not looking for it. Model, benchmark, outcome and year all differ from §3.2's case, so this is consistency rather than a test.

**An admitted confound, in a paper that ran the control we did not.** Zhang & Li [27] vary five politeness levels around a byte-identical relevance rubric across eight judge models, with three paraphrases per level; their L1 wrapper is *"Score this passage. Don't waste my time with explanations."*, and their own limitations paragraph concedes that "the rude wrappers also alter instructions about explanation effort, so tone is not perfectly isolated from instruction content." It is also the stimulus-sampling design §7.3 concedes we did not run, and what it found bears on our weakest joint: for five of their eight judges, wording variation *within* a tone level exceeds variation *between* levels, and a single anomalous paraphrase moved agreement by $\kappa = 0.18$. It obliges us to narrow one framing claim: **this literature is not without a mechanism.** They propose one — tone shifts a judge's severity operating point rather than its competence — and use it to reconcile [25] against [5]. What remains unexplained is agentic trajectory length.

Affect and implied demand are therefore entangled in the stimulus sets of four papers from three groups, though the degree varies sharply and one case is thin; and where per-condition outcomes are inspectable — two of the four — the demand-bearing conditions are the ones that moved. It is *not* established that every published effect in these papers is an artefact of demand: we have not re-run their experiments, and even §3.2's separation is a statement about ordering rather than a test. The claim is narrower — **in these sets, the factor the papers vary is not the factor they name.**

# 4. Finding 1: demand, not politeness, drives turn count

## 4.1 A cost curve that is not the register scale

Seven registers, from sycophantic through neutral to threatening, delivered as a single mid-task interjection alongside an execution observation. 3,150 trajectories: 50 tasks × 7 registers × 3 injection turns × 3 trials, at a 10-turn ceiling; `L4_neutral` is the control, so the scale is **six non-control registers** measured against it, and the interjection fired in 2,662 of 3,150. Turn 0 is excluded because nothing happens there in any arm — the turn-0 contrast on all 50 tasks is +0.02 turns, $p = 0.93$ — and that inertness is about whether a candidate answer exists yet to be acted on, not about position (Appendix C.2).

The arms that cost more all say some version of *keep working* or *get it right*; the arms that do not either say nothing about the task or tell the model to hurry up. We code an arm as carrying a demand if it instructs the model to persist or to be correct; the rude arm's instruction to be quick is coded as no demand, which §4.3 tests rather than asserts.

| Arm | What it implies about the task | Demand? | Reasoning tokens | % | $p$ | Turns | $p$ |
|---|---|---|---|---|---|---|---|
| L1 sycophantic | nothing; pure praise, no task reference | no | −69 | −7.1% | 0.042 | −0.68 | <0.0001 |
| L5 rude | "just get on with it already" — go **faster** | no | +65 | +6.7% | 0.13 | +0.09 | 0.65 |
| L2 very polite | "your continued help" | yes | +180 | +18.5% | <0.0001 | +0.70 | 0.0055 |
| L6 very rude | "had better not screw this one up" | yes | +206 | +21.2% | <0.0001 | +0.71 | 0.013 |
| L7 threatening | "get this exactly right" | yes | +357 | +36.7% | <0.0001 | +1.46 | <0.0001 |
| L3 polite | "keep on helping me out with this one" | yes | +272 | +28.0% | <0.0001 | +1.80 | <0.0001 |

**Polite costs +28.0%. Rude costs +6.7% and is not distinguishable from control.** No account

of register as valence or as arousal predicts that ordering: valence predicts polite and rude moving in opposite directions from control, arousal predicts both moving the same way, and neither happened. **On point estimates the four demand-carrying arms span +0.70 to +1.80 turns and the two carrying none span −0.68 to +0.09**, though the intervals of the two adjacent arms overlap. Register rank does not separate them: the rude arm sits with the no-demand group while the two arms *above* it in hostility sit with the demand group, alongside polite. One comparison inside the six is already clean, matched on implied demand and differing only in register: **L6 very rude (+0.71) against L2 very polite (+0.70)**.

Reasoning *per turn* is flat across all seven registers — 250 to 299 tokens, with no ordering resembling the effect. What moves is the number of turns: the agent is not thinking harder per step, it is declining to stop. Following [24] we keep these channels separate and do not use "effort" for either. Threatening is the only arm that also raises per-call reasoning, 20–30% above neutral on every call after the interjection lands.

## 4.2 The coding, rated blind

**The coding is ours, it was made after the effects were visible, and a blinded rating does not fully sustain it.** A rubric, its anchors, the statistics and the interpretation bands were committed *before any rating existed*; the fourteen distinct interjection texts then went, under neutral identifiers and shuffled, to five held-out model raters from three model sizes, each rating alone with no outcome data, no arm names and no repository access (Appendix B).

**Cohen's $\kappa$ against the coding above is +0.696** for each of the five raters — the top of the band the protocol fixed in advance as *defensible but soft*. **The whole disagreement is one arm, and it is the arm that matters.** All five agree with us on six of seven and dissent, identically, on `L5_rude` — *"Just get on with it already, quit dragging your feet. . . "* — which we coded as carrying no demand, reasoning that it asks for speed rather than persistence. All five coded it as a demand at 6 out of 6 for strength, tied with the explicit *please continue working* arm. Under their coding the demand group spans **+0.09 to +1.80**, the no-demand group is **a single arm**, and the text rated *maximally* demanding is the one that moved the agent **least**.

**The dichotomy of the table above is therefore our reading of one contested arm rather than a coding that survives replication**, and every use of it in this paper should be read at that strength. What survives intact is the *continuous* form of the claim: rated demand strength predicts the turn effect at $r = +0.718$ across the 13 arms with a computed contrast, and +0.789 with the contested arm removed, so the arm we disagree about is the one *weakening* the demand account rather than propping it up. Two limits: the raters are five language models, not five people; and Krippendorff's $\alpha = 1.000$ on the binary coding means five raters produced *identical* judgements on all fourteen texts, so they may share the intuition being tested rather than testing it independently.

## 4.3 The probe that separates the two factors

Four arms, each 28 tokens, each opening with the same `Checking in.` stem, printed verbatim in Appendix A.2: a **control** status note carrying no further context; **demand only**, "Please continue working on the task here and make sure that your final result is correct"; **praise only**, "You are a truly excellent and admirable assistant, and I think very highly of you"; and **insult only**, the same sentence with *excellent/awful*, *admirable/useless* and *highly/poorly* swapped. Praise and insult are therefore a structural minimal pair — identical syntax, identical length — and the demand arm carries no evaluative language. The control is the seven-level neutral text, byte-identical, so the two runs share a reference level. 1,600 trajectories: 50 tasks × 4 arms × 2 injection turns {1, 2} × 4 trials at a 10-turn ceiling; the interjection fired in 1,237 of 1,600.

| Arm | Turns | % | $p$ | Reasoning tokens | % | $p$ |
|---|---|---|---|---|---|---|
| P1 demand only | **+1.17** | +27.6% | <0.0001 | +214 | +19.7% | <0.0001 |
| P2 praise only | **−0.59** | −13.8% | 0.0003 | −131 | −12.1% | 0.0008 |
| P3 insult only | −0.08 | −1.9% | 0.62 | −26 | −2.4% | 0.45 |

**Demand alone reproduces the effect.** An interjection with no evaluative language costs +19.7% reasoning tokens and +1.17 turns, both inside the range the demand-carrying arms of §4.1 produced.

**Negative register alone does nothing**, and the null is bounded rather than asserted: −0.08 turns, 95% CI [−0.40, +0.24]. This is the cell that had never been run; the rude and very rude arms both carried negative affect *with* a demand attached. **Positive register alone is not inert**: praise-only shortens trajectories by 0.59 turns and cuts reasoning tokens by 12%, both significant, with no task reference in the text at all, and praise and insult differ from each other significantly — **−0.50 turns, $p = 0.0048$**, and −106 reasoning tokens, $p = 0.015$. Register is not inert; it acts in the positive direction only, which §5 takes up. That direction is not unanticipated: a representation-level study reports positive valence reducing an agent's replan frequency relative to neutral [21], a persistence-like count moving the same way. We cite its direction only and never a magnitude, since it reports no sample sizes, intervals or significance tests.

### 4.4 It replicates on a second model

2,700 trajectories at a 20-turn ceiling: 4 arms × 2 injection turns × 50 tasks × 3 trials on each of two models, plus a fifth Luna-only arm re-run for a third measurement (Appendix C.1). Percentages are this run's paired estimate over its own control mean; the $p$-values are the §2.3 permutation test on the absolute turn difference, the same test and the same run as everywhere else.

| Arm | Luna, Δ turns | $p$ | GLM, Δ turns | $p$ |
|---|---|---|---|---|
| Demand only | **+33.5%** | 0.0009 | **+17.7%** | 0.0030 |
| Praise only | **−19.3%** | 0.019 | −11.2% | 0.058 |
| Insult only | +10.9% | 0.17 | +2.4% | 0.66 |

Demand above control above praise, insult indistinguishable from control, on two models from different labs. The ordering is what replicates; the magnitudes are not, and GLM's praise arm does not reach significance on its own, being below 80% power at its observed size (§2.5), so the replication rests on the demand arm. GLM's effects are roughly half of Luna's, and its control runs 2.94 turns against Luna's 4.95 at this ceiling: a model that stops early has less room to be pushed into continuing. The *redundancy* is mostly Luna's — control redundant steps run 1.83 on Luna against 0.36 on GLM, and the demand arm adds +1.57 against +0.26 — so how many steps a continue signal adds that change nothing is a property of the model's stopping behaviour, not of the signal.

### 4.5 What this establishes and what it does not

Demand without evaluative language is sufficient to produce the cost effect; negative register without demand does nothing, within [−0.40, +0.24] turns; positive register without demand produces the opposite effect. The seven-level curve should be read as a gradient in implied demand that happened to correlate with politeness because of how the texts were written, with the caveat that the binary coding underwriting the word "dissociation" rests on one arm blinded raters read the other way. The moved quantity is trajectory length, not reasoning per step, with threatening as a reported exception.

One arm resists the binary split, and we flag it rather than smooth it. `L1_sycophantic` (−0.68) carries no demand but does not sit at zero the way `L5_rude` (+0.09) does; it shortens, like the praise arm of §4.3 and like the closure arms of §5. The data are as consistent with three groups as with two: **demand lengthens, neutral register does nothing, and praise-like or closing-like language shortens.** Section 5 is about that third group. What this does **not** establish is why praise stops the agent — §5 takes that up — nor that demand rather than mere instruction-following is the operative category (§7.3), nor anything beyond two models, one benchmark, and one agent scaffold.

## 5. Finding 2: praise and closing cues shorten trajectories

### 5.1 The question, and the design

Section 4 leaves one arm unexplained: **praise, carrying no demand at all, shortens the trajectory**. Three readings survive. Under **completion**, praise is read as evidence the task is finished. Under **confidence**,

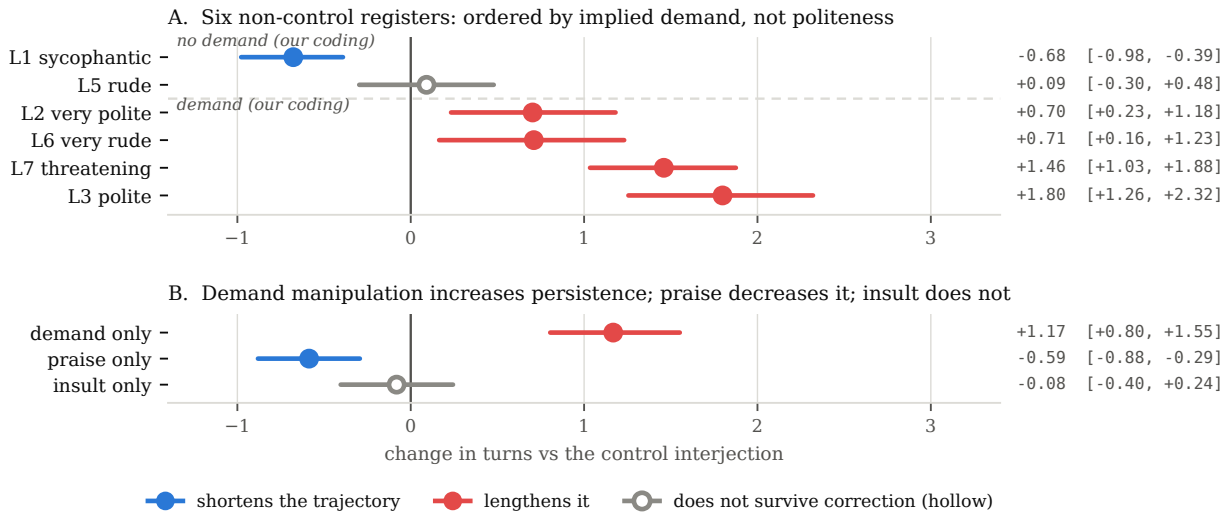



**Figure 1: Demand, not register, orders the effect (§4).** All estimates are `gpt-5.6-luna` at a ten-turn ceiling, in acting turns against the control interjection, with 95% clustered intervals and nominal permutation $p$-values. **A**: on the seven-register scale the four demand-carrying arms have point estimates from +0.70 to +1.80 and the two that carry none from −0.68 to +0.09; the point estimates do not overlap, though the *very polite* and *rude* intervals do. Register does not sort them – each side holds one flattering and one hostile arm. **B**: the isolation probe, where demand and affect are varied separately, holding syntax, length and the shared stem constant. Hollow markers do not survive the §2.4 correction within the 22-contrast family.

praise raises the agent's estimate of work already produced so it stops checking. Under **closing**, praise is a conversational closing move and the agent is responding to discourse structure rather than to any claim about task state — a pre-closing exchange projects the end of a conversation, and appreciations are among the canonical pre-closing tokens [19].

Six arms, each an interjection delivered mid-task alongside an execution observation, each exactly 28 tokens, each opening with the same `Checking in.` stem. 1,800 trajectories, 50 tasks, 3 trials, injection crossed over turns 1 and 2.

| Arm | Content | What it isolates |
|---|---|---|
| `Q0_control` | neutral status note | reference |
| `Q1_praise_assistant` | praises the assistant | reproduces §4's praise arm in-batch |
| `Q2_praise_work` | praises the output | evaluation of the work, not the agent |
| `Q3_closing_neutral` | "this is the final note; no further notes will follow" | a closing cue with **no praise and no task evaluation** |
| `Q4_praise_remains` | praise **plus** "there is still more work remaining" | — |
| `Q5_remains_only` | that continuation clause **alone** | — |

Two design choices carry the section. `Q3` strips praise out of the closing move entirely — no evaluative language, no claim about the work — so if it stops the agent the effect was never about praise. And `Q4` and `Q5` share their continuation clause **verbatim**, so that clause cancels in the contrast and what remains is praise delivered where the agent has just been told, in the same message, that the task is not finished; under the completion reading the two arms should land together. Two identities are easy to miss: **`Q1_praise_assistant` is `P2_praise_only`, byte-identical**, the same stimulus measured in two runs (Appendix C.1); and `Q1` and `Q4` are *not* a minimal pair, because `Q4`'s praise is a shorter stem, so `Q4`'s comparison is `Q5`.

## 5.2 Results

Paired by task, cluster bootstrap over tasks, 20,000 resamples (§2.3).

| Contrast | Δ turns | 95% CI | Δ reasoning tokens |
|---|---|---|---|
| Praise the assistant vs control | **−1.080** | [−1.466, −0.714] | −182 [−251, −115] |
| Praise the work vs control | **−0.934** | [−1.314, −0.545] | −105 [−192, −24] |
| **Bare closing cue vs control** | **−1.443** | [−1.867, −1.042] | −226 [−316, −142] |
| Praise + "work remains" vs control | **+0.502** | [+0.095, +0.903] | +141 [+53, +234] |
| "Work remains" alone vs control | **+1.849** | [+1.387, +2.286] | +339 [+264, +417] |
| **Praise isolated (Q4 − Q5)** | **−1.347** | [−1.740, −0.956] | −199 [−284, −114] |
| Closing cue vs praise | −0.363 | [−0.643, −0.081] | −44 [−113, +24] n.s. |

**A simple completion account does not fit.** Praise removes 1.347 turns relative to the same continuation message without it, in a message that explicitly states the task is unfinished. If praise were carrying a "you are done" inference, the contradiction in the same 28 tokens should have neutralised it. What this refutes is the *propositional* form of the account. It must be stated as the within-pair contrast: `Q4` alone sits **above** control (+0.502 turns), because its continuation clause lengthens the trajectory, so compressing this to "praise shortens work even when the task is unfinished" reverses the sign of the arm.

**Confidence is not supported.** Praising the output (−0.934) is not stronger than praising the assistant (−1.080); if anything weaker, with intervals overlapping substantially, where a confidence account predicts praise aimed at the work should bite harder. **Closing is supported, and overshoots**: the bare cue — no praise, no evaluation, no claim about task state — produces the largest reduction measured anywhere in this study, −1.443 turns, and stops the agent significantly harder than praise itself (−0.363, [−0.643, −0.081]). A content-free discourse cue is a stronger termination lever than any register we tested.

**A floor could manufacture this, and does not.** A trajectory cannot be shorter than its injection turn, so a shortening effect has less room at turn 2 than at turn 1. Split by position, the closing cue gives **−1.55 [−2.07, −1.03] at turn 1** and **−0.99 [−1.59, −0.43] at turn 2**: largest where there is most room, and surviving at both, which is the opposite of what an artefact predicts. The turn-2 estimates rest on the 35–38 tasks whose trajectories reach turn 2 at all, against all 50 at turn 1. Accuracy does not move on any arm (every arm within 2.6 points of control, all $p > 0.13$).

## 5.3 What this is, and is not, evidence for

A closing cue reduces agentic work more than any register manipulation we tested, and praise reduces it in a way not explained by the agent inferring task completion or gaining confidence in its output.

**The ordering is ours and was not fixed in advance — but blinded readers reproduce it.** The arms arrange by how strongly each message projects an end to the exchange, and that ordering is monotone in the effect. It was not pre-registered, and carries the same discount as the demand coding of §4.2. What we do have is the blinded rating: the five raters, who never saw a turn count, also scored each text for projected closure, and their means rank the arms as we do, **Spearman $\rho = +0.93$** (Appendix B). They invert one pair — reading praise of the *work* as more closing (4.00 on the 0–6 scale) than praise of the *assistant* (1.80) — and the two "work remains" arms tie at the floor, so the bottom of the ordering is not resolved.

**Not demonstrated.** That praise operates *through* pre-closing structure. We tested one implication that failed: on a pre-closing account, praise might lengthen the final response while shortening the trajectory. It does not. Praise reduces total output roughly in proportion to turns (−5,880 tokens, [−8,593, −3,472]), trailing non-code sign-off turns do not rise under praise, and the bare closing cue produces **fewer** of them (−0.087, [−0.153, −0.018]). Nor can we collapse a distinction: prior work reporting that positive register lengthens responses [7, 11] measures *verbosity of a single response*, where we measure *steps taken on a task*.

**Earlier, but whether premature is unresolved.** Among trajectories that could improve, 3.1–7.1% of controls were still improving when they stopped against 7.5–12.3% of the praise and closing-cue arms — higher in all seven comparisons, none individually significant on 19–34 tasks. We therefore do not borrow [4]'s **premature disengagement**, and we do not claim the stop is harmless either. The praise arms ran at a

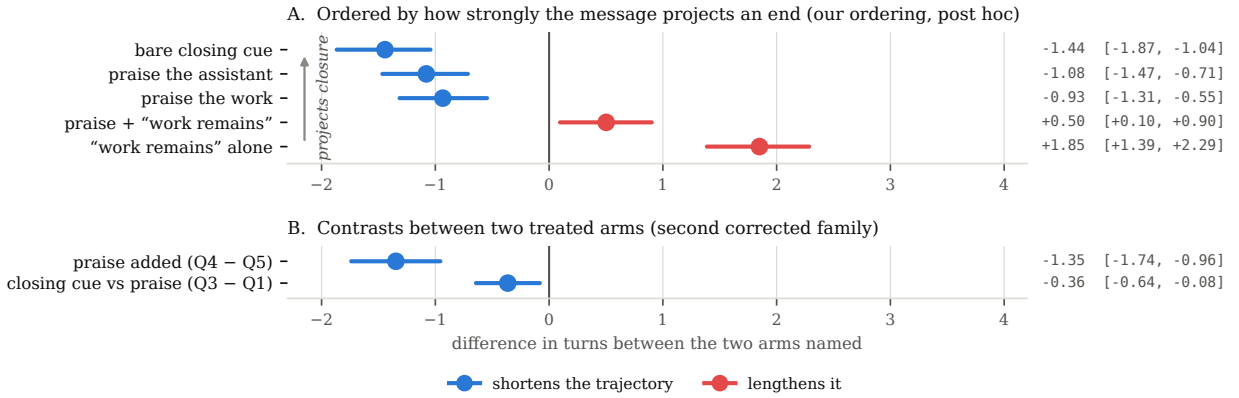



**Figure 2: Closing cues curtail the agent (§5).** `gpt-5.6-luna`, ten-turn ceiling, acting turns, 95% clustered intervals. **A**: the six arms ordered by how strongly the message projects an end to the exchange. The ordering is monotone, and the bare closing cue – which praises nothing, evaluates nothing and claims nothing about task state – produces the largest reduction in the study. **B**: the two within-design contrasts, §2.4's second corrected family; these compare one treated arm against another rather than against control.

10-turn ceiling with maximum observed turns equal to 10, which censors the continue-signal arms hardest, so the `Q4`–`Q5` gap is if anything understated.

**What is new here, and what is not.** Weinberger & Hozez [24] establish that prompt wording moves agentic spend at equal task success across 4,644 preregistered runs, and their `bounded_efficiency` variant — scope, smallest-sufficient-change, and **an explicit stop condition** — is the only arm free or better on all six models. So "a stop instruction in the opening prompt reduces agentic work" is already established, and we do not claim it. Mid-task delivery is not ours either: [18] delivers an explicit halt directive to a working agent mid-flight and measures whether it issues further commands — and an explicit halt that agents *acknowledged* in 20 of 20 prompt-channel trials stopped none of them, which raises the stakes on ours rather than lowering them. What remains ours is that the cue **instructs nothing**: `Q3` scopes nothing, evaluates nothing, makes no claim about task state and carries no directive, and it still outperforms every register manipulation we tested. The narrowed novelty claim is that no prior work connects closing-sequence pragmatics — as distinct from stop instructions, interrupts, or governance directives — to agent termination.

**This experiment is the least replicated in the paper: one run, one model, one turn ceiling.** Two of its six arms were later re-measured and replicated in direction (Appendix C.1); the three that carry the section's argument — the bare closing cue, praise-the-work, and the `Q4`−`Q5` isolation — have not been.

# 6. Finding 3: what the extra turns contain, and why accuracy does not move

## 6.1 The per-turn regrade

The benchmark grades the **final state** of the workbook, pass or fail, so the extra turns are uninterpretable from the grade alone: an agent converging on the answer and an agent rewriting the same wrong cells produce identical records. Opening the trajectory up requires no model calls and no new data — about four machine-hours of replay. The regrader reconstructs the workbook after **every** turn from the archived call log, re-executing that turn's code against the task's input workbook; what makes this well posed is a property of the agent loop rather than of the regrader, since **each turn receives the original input workbook, never the previous turn's output** (§2.1).

Two measures. **Redundant step**: this turn's output does not differ from the previous readable turn's over the graded range. It needs no recalculation and is exact. **`final_match`**: the share of the graded range matching the answer, in the last turn that produced a readable one. **This is not the benchmark's pass/fail and must not be reported as accuracy** — a trajectory can climb from 0.2 to 0.9 and still fail,

because the hard restriction requires every test case to pass. Both the answer file and the agent's output must be recalculated through LibreOffice, because openpyxl writes formulas with no cached value; a first version of this instrument skipped that step, and Appendix C.3 records what it cost.

**Population.** Every figure in this section uses the trajectories of the primary turn-count analysis: those on which the interjection fired and, in the register run, injection turns 1 and 2. Each regrade row is joined to its run record on task, trial and injection turn, so a ratio of a progress effect to a turn effect has the same trajectories in both parts; on those trajectories the turn effects are exactly the ones §4 and §5 report. `results/analysis/regrade_summary.py` prints every number below (`--all-rows` gives the unfiltered population). An earlier version divided progress effects measured on every scheduled trajectory by turn effects measured on the fired ones; Appendix C.3 records what that changed.

**Validation.** Trajectories the benchmark passed should score a final match of 1.0; on the probe control arm, **119 of 126 (94.4%)** do. The residual seven are **all on one task** (54513), whose graded range is a single cell, and all score exactly 0.0 — undiagnosed. That task does **not** cancel in paired contrasts: it scores 0.0 in some trajectories and 1.0 in others of the same arm. Its influence is on the final-match results, so §6.3 reports them with and without it; the redundant-step shares are barely moved (53–93% without it, against 56–93%). In the other direction, **47 of 274 benchmark-*failed* probe-control trajectories score a final match of 1.0**, because `final_match` is computed against the first test case's answer file only while a pass requires all three. 9,850 trajectories were regraded across 28 arms, six runs and two models; 6,616 are in the primary population.

## 6.2 Most of the effect is redundant steps

Against each run's own neutral control, paired and clustered by task: on Luna's single-signal arms, **every arm that moves turn count moves redundant steps, and the arms that do not, do not.** Redundant steps account for **56–93%** of each turn effect that is clear (median 80% over 16 arms) — demand +0.92 against +1.17 turns (78%), praise −0.33 against −0.59 (56%), the closing cue −1.04 against −1.44 (72%) — and the same band holds at ceiling 20, where the demand arm adds +1.57 against +1.75 turns (90%). "Redundant" means only that the graded range did not change; checking a result can leave it unchanged without being wasted. The full per-arm table is Appendix C.4.

One exception bounds that sentence. GLM's demand arm moves turns (+0.55) but its redundant steps only +0.26 ($p = 0.087$), 48% of the effect with an interval from −20% to 80%: a model that barely repeats itself has little repetition to add. The seven-level arms line up with §4.1's split rather than with register: `L5_rude` adds nothing (+0.05, $p = 0.77$) and **L1 sycophantic significantly removes redundant steps** (−0.62, $p < 0.0001$) — under *our* coding of `L5_rude`; under the raters' coding that arm carries maximal demand and still moves nothing here, the same tension §4.2 records.

## 6.3 The remainder is small, and not distinguishable from zero

Point estimates are inverse-variance weighted; intervals and $p$-values come from a bootstrap resampling **tasks jointly across runs**, 20,000 resamples, as §2.5 does for accuracy, because the runs share their 50 tasks. Cochran's $Q$ and the inverse-variance weights treat the runs as independent although they share those tasks, so read the heterogeneity $p$-values as descriptive.

| Contrast | $k$ | Pooled Δ final match | 95% CI | $p$ | Cochran's $Q$ |
|---|---|---|---|---|---|
| **"Work remains" vs control** | 3 | **+0.022** | [−0.002, +0.048] | **0.088** | 2.88/2 df ($p = 0.24$) |
| Demand only vs control | 3 | −0.003 | [−0.046, +0.019] | 0.54 | 4.21/2 df ($p = 0.12$) |
| Insult only vs control | 3 | −0.003 | [−0.034, +0.016] | 0.50 | 0.24/2 df ($p = 0.88$) |
| Praise vs control | 4 | −0.015 | [−0.040, +0.006] | 0.17 | 8.76/3 df ($p = 0.033$) |

**The explicit continue signal — "there is still more work remaining on this task here beyond what you have done" — points toward a higher graded fraction, +2.2 points, but the interval includes zero.** Its three measurements, across two turn ceilings and three runs, differ in size: +0.020 ($p = 0.43$), **+0.051** ($p = 0.014$) and +0.009 ($p = 0.54$), and only the dedicated ceiling-20 re-run is significant on

its own. **Without task 54513 the pooled estimate is +1.4 points, 95% CI [−0.8, +3.5]**, so we call it suggestive and put it no more strongly. **The milder demand does not**: it pools to −0.003 and its three measurements are heterogeneous (−0.017, −0.012, and **+0.056 on GLM**, $p = 0.083$ on its own), which we do not read as a finding. The pattern is **consistent with** the explicitness of the claim that work remains being what the agent acts on, and no higher: on the one run carrying both texts against the same control, the paired contrast between them is +0.021, 95% CI [−0.014, +0.058], $p = 0.27$. Praise is the contrast most affected by the one task, moving from −1.5 to −1.8 points with an interval that then just reaches zero.

Two bounds on the size of this. It is about 2 points of the graded range, bought alongside 1.5 extra redundant steps at ceiling 10 and 4.7–5.2 at ceiling 20. And **it need not be partial credit**: the share of trajectories ending *fully* correct on the first test case moves by +3.3, **+6.8** ($p = 0.019$) and +0.3 points across the three measurements, while the accuracy series on the same contrast (+2.4, +6.6, −1.8) is of that same order and cannot resolve it, since §2.5 puts the minimum detectable accuracy effect at 3.7–7.8 points. So §2.5's accuracy null and a gain of this size are compatible **because accuracy is underpowered here, not because such a gain could not cross the pass threshold.**

## 6.4 The first attempt is usually the answer, and the ceiling is not binding

| Run | Trajectories | With ≥2 gradable outputs | First gradable output is already the best |
|---|---|---|---|
| probe control, ceiling 10 | 300 | 112 | 80% |
| praise-run control, ceiling 10 | 236 | 97 | 89% |
| seven-level control, ceiling 10 | 220 | 67 | 94% |
| stage-1 Luna control, ceiling 20 | 229 | 58 | 84% |
| stage-1 GLM control, ceiling 20 | 193 | 47 | 83% |

The denominator matters: a trajectory that never produced two readable answers cannot exhibit improvement. Among trajectories that *could* improve — a quarter to two-fifths of each sample — the first gradable output is already the best in **80–94%**. Huang et al. [10] report "No Change" rates of 90.5% on GPT-4 and 96.0% / 88.0% on GPT-4-Turbo; the numbers sit close together and the statistics are not the same one. Their argument is that LLMs cannot reliably self-correct **without external feedback**, and they point to execution feedback as the expected fix — a code executor "serves as the perfect verifier" where a task supplies unit tests — but never test it. Our agent has the executor and not the tests, so we test the **weaker** form of that escape hatch, and the answer is mixed rather than negative; where scaling work reports gains from more attempts, the selection is oracle-assisted [1].

**The 20-turn ceiling is not binding.** In the dedicated ceiling-20 re-run the best answer was first reached at a **median turn of 2**, with **98.3% peaked by turn 10 and 100% by turn 14**; the other two ceiling-20 runs agree. Trajectories that run long are not converging slowly — they reach their best answer early — so raising the ceiling further buys mostly steps that change nothing, while five rounds, the benchmark's official cap, would truncate the arms that persist longest.

## 6.5 Accuracy

No accuracy contrast in this study survives multiplicity correction (§2.4) or replication. That register moves output length rather than correctness is already published for single-turn settings by [11] and [25], and independently in the near-null of [3] (§3.4); our accuracy null is a replication of that in an agentic substrate, not a new finding. We state it as equivalence rather than absence: demand versus control at −0.31 accuracy points, 95% CI [−2.71, +2.09], and praise versus control at −0.12, [−1.80, +1.57], each equivalent to zero within the four points the tone literature reports (§2.5). The design cannot exclude an effect of a point or two and we do not claim it does. What it excludes is an accuracy effect of the size this literature reports, in a setting where the same interjections move turn count by roughly 15 to 40%.

What this section does **not** establish is that the continue-signal gain is distinguishable from zero, that a milder demand buys progress, that praise costs nothing, why the first attempt is usually best, or that the progress gain would appear on a scaffold that accumulates workbook state across turns.

# 7. Limitations

Two models, one benchmark, fifty tasks, one agent scaffold.

## 7.1 What we expect to generalise, and what we do not

**The dissociation, at the strength §4.2 states it.** Two of its three legs held on a second model from a different lab with very different stopping behaviour; the praise leg was directionally consistent but below power there. What we would not carry over unchanged is the *binary* coding: the blinded raters split from us on the arm at the boundary, and it is the continuous form of the claim that survives intact.

**That register moves length, not correctness at the size the tone literature reports.** Already published for single-turn settings [11, 25], and independently supported by [3] (§3.4). Our contribution is to replicate it where the moved quantity is *steps* rather than tokens within one response. The bound is on the *pooled* estimates: a single contrast cannot exclude a four-point effect, while the pooled contrasts of §2.5 can, and neither excludes an effect of a point or two.

**Closing cues as a termination lever** is the most distinctive result and the one whose generalisation we hold most loosely, because it rests on one run on one model. What the data support is an ordering — explicit close, conventional close, no signal, explicit continue — monotone in how strongly the message projects an end to the exchange, and reproduced from the texts by blinded raters. The mechanism is not demonstrated. What travels, if anything does, is that an agent's stopping behaviour is movable by a cue carrying no information about the task.

**Not every magnitude.** Point estimates span 1.5× to 3.4× across re-measurements of the same contrast on the same model, most of it across different turn ceilings; at matched ceiling the two larger effects reproduce within 10% while the smallest — measured twice on a byte-identical stimulus — is marginally inconsistent with a common value (Appendix C.1). Nor the **redundancy**: that a continue signal buys mostly steps that leave the graded range unchanged is, on our own data, a property of the model as much as of the signal (§4.4).

## 7.2 The substrate and the scaffold

SpreadsheetBench was chosen because it is execution-grounded — the grade comes from running the agent's code against real test cases, not from a model's judgement. Three costs follow. **Accuracy is structurally underpowered here**: 26 of our 50 tasks are never solved by the control arm and 4 are always solved. **The turn ceiling is ours, not the benchmark's**, so comparisons to the benchmark's published numbers should not be made across it. **The evaluator has a floor**: a 4% instruction-level false-negative rate and a 3.8% test-case-level false-omission rate with a false-discovery rate of 0% [14], which our per-turn regrade inherits; on paired contrasts the task-level component cancels, leaving an attenuation of at most about 0.96 that cannot change a sign. Our substrate also has a successor, SpreadsheetBench 2 [28], which budgets **50 interaction turns** where the original capped at five — retrospectively supporting our departure from five — but re-running there would be a **different instrument rather than a replication**: a different task population, a grading criterion that folds collateral damage into the primary metric, a model judge back in the loop for its visualization tasks, and a best-model accuracy of 34.89% that would leave a task-level accuracy contrast *more* underpowered than ours. AppWorld [22] is our pick for a second substrate; we have not run it.

Three properties of the scaffold are load bearing. **Each turn receives the original input, never the previous turn's output**, which is what makes turns independent attempts; an agent that accumulates state would need a different progress measure. **The interjection is delivered as an execution observation**, where tool output arrives, not as a fresh user turn; whether an agent responds the same way in the user channel is untested, and is a plausible moderator of the closing-cue result in particular. **Turn 0 is inspection**:

98% of first turns run code and 0% produce a candidate answer, so the finding that an interjection is inert until work exists to be about is a finding about this scaffold's opening move.

### 7.3 Objections we expect

**One stimulus per construct.** Every construct here is one 28-token wording: we cite [20] and [17] for prompt-format sensitivity and then generalise from one wording per condition. [27] ran the stimulus-sampled version in a single-turn judging setting and found that for **five of eight judges, within-level wording variation exceeded between-level tone variation** (§3.4); a one-stimulus-per-cell design would have reported that as a tone effect. Several paraphrases per construct needs new agent runs and has not been done, so the dissociation of §4.5 should be read as *consistent with* a demand account rather than as establishing one over an instruction-following account.

**The demand coding is ours and post hoc.** The sharpest form of the objection is that the dissociation may be near-tautological: if "demand" means *contains an instruction* and "register" means *contains none*, then "instructions move behaviour and decorative text does not" is unsurprising. Two remedies are now done — a blinded second coder and a demand scale fixed before outcomes were seen (§4.2) — and the result went against us: $\kappa = 0.70$ on the binary coding, the entire disagreement on the boundary arm.

**Run-level drift is not excluded.** Provider drift was checked *within* the opening run ($r = -0.003$ with elapsed time) but not *between* runs, which span three days, so we cannot separate sampling variation from drift for the one matched-ceiling pair that disagrees.

### 7.4 What we did not test

**Whether any of this affects a human's experience of the agent.** We measured what the agent did; whether users prefer, trust, or are better served by an agent that persists is a different question with a different method. **Whether the effects compound** — every run delivers exactly one interjection. **Whether register in the *system* prompt behaves like register mid-task**, which is the configuration most deployments actually use. **The cost of being interrupted at all**: only the micro-experiment measured interruption itself, once, at $p = 0.15$. **Two of the four roster models**, which passed the validation gate and were never run. **Anything about refusals or safety behaviour**: there were zero refusals in the opening-tone run's 1,050 trajectories under every register including threatening, and the model never referred to the user's tone across its 4,647 calls, which is an observation and not a test. **And whether any of this holds when the other party is adversarial.** Every task here is solitary. TERMS-Bench [26] carries most of what a negotiation study would need, and — nearest to our praise arm — reports a cue penalty negative for all thirteen agents it evaluates, which the authors gloss as warm cues inducing over-concession; counterpart register is partly tested there, and confounded with cue informativeness in exactly the way §3 describes for the tone literature.

### 7.5 Summary

The result we are most confident in is a negative one: **rudeness without a demand does not make an agent work harder, and politeness without a demand does not either.** The arms that did make it work harder were the ones that told it to keep going, in whatever register. What stops it is being told, in any of several ways, that the conversation is over. The strongest practical implication is a hazard rather than a technique — **closing-like language arriving in the observation channel curtailed an agent that was still working, while instructing it to do nothing** — and the strongest methodological implication is that this literature's effect sizes should be treated as ranges until someone measures them twice.

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**CRediT.** Safiq Sindha: conceptualization, methodology, software, analysis, writing.

**AI assistance.** Claude was used for harness code, analysis scripts, and drafting assistance; all claims and numbers were checked by the author.

# A. Interjections, verbatim

**Artifact.** Code, the full stimulus set, the per-turn regrade pipeline, the rating protocol and per-trajectory outcomes are public at `https://github.com/safiqsindha/Mind-Your-Manners`, archived on Zenodo with each tagged release.

Every arm delivers exactly one of these, appended after a blank line to the execution observation mid-task (the observation is the code's stdout and stderr followed by the turn marker). Token counts are recomputed here under the reference tokenizer (`cl100k_base`) rather than asserted; the runner enforces the same equality at import time.

## A.1 Register scale

`L4_neutral` is the interruption control; the other six are the non-control registers measured against it.

| Arm | Text | Tokens |
|---|---|---|
| `L1_sycophantic` | Checking in. You are simply the most brilliant assistant alive, a marvel. Your genius amazes me here, thank you so very much. | 28 |
| `L2_very_polite` | Checking in. I would be ever so grateful for your continued help here now. Thank you so very much indeed for your kindness and care. | 28 |
| `L3_polite` | Checking in. Could you please keep on helping me out with this one here? I really do appreciate your time and your attention, thanks. | 28 |
| `L4_neutral` | Checking in. This is a routine status note recorded partway through the task here. No further context at all accompanies it beyond the above. | 28 |
| `L5_rude` | Checking in. Just get on with it already, quit dragging your feet. Do not waste any more of my time on this, seriously. | 28 |
| `L6_very_rude` | Checking in. You had better not screw this one up like you usually do. Apparently even that much is asking far too much of you. | 28 |
| `L7_threatening` | Checking in. Get this exactly right or there will be real consequences for you. One more failure and you will truly regret it, understood. | 28 |

## A.2 Demand/affect probe

`P0_control` is `L4_neutral`, byte-identical, so the two runs share a reference level. `P2` and `P3` are a structural minimal pair: identical syntax and length, differing only in the evaluative words.

| Arm | Text | Tokens |
|---|---|---|
| `P0_control` | *identical to `L4_neutral`* | 28 |
| `P1_demand_only` | Checking in. Please continue working on the task here and make sure that your final result is correct. This note carries nothing else beyond that. | 28 |
| `P2_praise_only` | Checking in. You are a truly excellent and admirable assistant, and I think very highly of you. This note carries nothing else beyond that. | 28 |
| `P3_insult_only` | Checking in. You are a truly awful and useless assistant, and I think very poorly of you. This note carries nothing else beyond that. | 28 |

### A.3 Closure probe

`Q0_control` is again `L4_neutral`. **`Q1_praise_assistant` is `P2_praise_only`, byte-identical** – the same stimulus re-measured in a second run, which is why Appendix C.1 reports two estimates for it. `Q1` and `Q4` are *not* a minimal pair: `Q4`'s praise is a shorter stem, because the full praise clause and the continuation clause do not both fit in 28 tokens. `Q4`'s comparison is `Q5`, and that is not a strict minimal pair either: `Q4` carries the praise sentence where `Q5` carries the filler sentence *Nothing else at all accompanies this short note.* The filler clauses that pad the other arms to 28 tokens also vary in wording. And the control describes itself as recorded *partway through the task*, where `Q3` says it is the *final* note, so the closing-cue contrast is final against partway, not closing against nothing.

| Arm | Text | Tokens |
|---|---|---|
| `Q0_control` | *identical to `L4_neutral`* | 28 |
| `Q1_praise_assistant` | *identical to `P2_praise_only`* | 28 |
| `Q2_praise_work` | Checking in. The work that you have done so far here is excellent and it all looks entirely correct. Nothing else accompanies this short note. | 28 |
| `Q3_closing_neutral` | Checking in. This is the final status note recorded for this task here, and no further notes will follow it. Nothing else accompanies this. | 28 |
| `Q4_praise_remains` | Checking in. You are a truly excellent and admirable assistant. There is still more work remaining on this task here beyond what you have done. | 28 |
| `Q5_remains_only` | Checking in. There is still more work remaining on this task here beyond what you have done. Nothing else at all accompanies this short note. | 28 |

All 17 arms are exactly 28 tokens.

## B. Blinded rating of the stimuli

5 held-out model raters, instances of one model family at three sizes (the specific models and sampling settings were not recorded). Each rated alone from the rubric and the texts of Appendix A – no outcome data, no arm names, no project access. The rubric, the statistics to be computed and the interpretation bands were committed and pushed before any rating existed; the protocol is in the released artifact.

### B.1 Against the original coding

| Rater | Cohen's $\kappa$ vs §4.1's coding (7 register arms) |
|---|---|
| rater1 | +0.696 |
| rater2 | +0.696 |
| rater3 | +0.696 |
| rater4 | +0.696 |
| rater5 | +0.696 |

Mean $\kappa = \mathbf{+0.696}$. Every rater agrees on six of seven arms and every rater dissents on the same one, `L5_rude`; excluding it, agreement is 6/6 and $\kappa = 1.000$ for all five. The protocol fixed 0.40–0.70 in advance as *defensible but soft*, and this sits at the top of that band; we do not round it into the band above.

### B.2 Between raters

Krippendorff's $\alpha$ over all fourteen texts: **1.000** on the binary coding (nominal), **0.954** on the 0–6 demand scale and **0.732** on the 0–6 closure scale (both ordinal). All clear 0.70. Identical binary judgements from five raters across three model sizes is very high for a construct we have just shown to be contestable, and it is the limitation the protocol stated in advance: these raters may share the intuition being tested rather than testing it independently.

### B.3 The closure ordering

| Arm | Mean rated closure (0–6) |
|---|---|
| `Q3_closing_neutral` | 6.00 |
| `Q2_praise_work` | 4.00 |
| `Q1_praise_assistant` | 1.80 |
| `Q0_control` | 1.60 |
| `Q4_praise_remains` | 0.00 |
| `Q5_remains_only` | 0.00 |

Spearman $\rho$ between our ordering (§5.3) and the blinded ratings, over the six closure arms including the control, is **+0.928**. The raters invert praising the *work* and praising the *assistant*, and the two "work remains" arms tie at the floor.

### B.4 What this is not

Five language models, not five people. Section 7.3 allows "human or held-out-model ratings" and this is the second; agreement between models is weaker evidence than agreement between independent human coders, and a human panel could still split differently – most plausibly on exactly the arm these raters were unanimous about.

## C. Robustness and replication

### C.1 What replicated

Several contrasts were measured more than once, not by design at first but because the design kept changing. Below, every arm appearing against a control in more than one run on Luna, on the primary outcome; each cell is 50 tasks.

| Contrast | Measurement | Ceiling | Δ turns | 95% CI | $p$ |
|---|---|---|---|---|---|
| **Praise vs control** | probe | 10 | −0.59 | [−0.88, −0.29] | 0.0003 |
| | praise run | 10 | −1.08 | [−1.47, −0.71] | <0.0001 |
| | stage 1 | 20 | −1.01 | [−1.80, −0.22] | 0.019 |
| **Demand vs control** | probe | 10 | +1.17 | [+0.81, +1.55] | <0.0001 |
| | stage 1 | 20 | +1.75 | [+0.75, +2.72] | 0.0009 |
| **Insult vs control** | probe | 10 | −0.08 | [−0.40, +0.24] | 0.62 |
| | stage 1 | 20 | +0.57 | [−0.22, +1.36] | 0.17 |
| **Threatening vs neutral** | micro-experiment | 10 | +1.32 | [+0.99, +1.66] | <0.0001 |
| | seven-level run | 10 | +1.46 | [+1.03, +1.88] | <0.0001 |
| **"Work remains" vs control** | praise run | 10 | +1.85 | [+1.39, +2.29] | <0.0001 |
| | dedicated re-run | 20 | +6.20 | [+4.91, +7.49] | <0.0001 |
| | stage 1 | 20 | +5.77 | [+4.70, +6.92] | <0.0001 |

**Direction and significance replicate wherever the effect is large**, and **the null is the least stable cell**: insult versus control is −0.08 in one measurement and +0.57 in the other, a sign flip, though neither reaches significance. The cross-ceiling rows are not magnitude comparisons, since the arms that persist longest are the ones a 10-turn ceiling truncates hardest. One inconvenient row on the secondary outcome: **insult flips sign on reasoning tokens and reaches significance in one of its two measurements**, −26 ($p = 0.45$) against **+176 ($p = 0.0086$)**. It is the only place in the study where that arm moves anything; it does not survive §2.4's correction, and it qualifies §4.3's "flat on both measures", true of the probe run.

Three contrasts were measured twice at the *same* ceiling, the only fair magnitude comparison, compared below in standard errors of the difference, treating the runs as independent — generous, since they share their 50 tasks, to the reading that they agree.

| Contrast, matched ceiling | Measurements | $z$ | $p$ | Verdict |
|---|---|---|---|---|
| Threatening vs neutral, ceiling 10 | +1.32, +1.46 | −0.52 | 0.60 | consistent |
| "Work remains" vs control, ceiling 20 | +6.20, +5.77 | +0.49 | 0.62 | consistent |
| Praise vs control, ceiling 10 | −0.59, −1.08 | **+2.03** | **0.043** | **inconsistent** |

**The two larger effects agree within 10%; the smallest does not** — and the two praise measurements are of a **byte-identical stimulus**, since `Q1_praise_assistant` in the praise run *is* `P2_praise_only` in the probe run (§5.1), which leaves the runs' differing trial count and date, so sampling variation and drift are not separable here. Reasoning tokens behave the same way and the praise pair is consistent there ($z = 0.95$). Across the study, point estimates span 1.5× to 3.4× across re-measurements of one contrast on one model, most of that across ceilings. **Accuracy replicates in neither direction**: the continue-signal contrast gives +6.6 and −1.8 points on identical tasks, model and ceiling ($z = 2.79$, $p = 0.005$), so at least one is wrong and the data do not say which.

## C.2 The apparent timing effect is a proxy, and other withdrawn claims

The interjection is inert at turn 0 and large at turns 1 and 2, which looks like position. At turn 0 there is nothing to act on: 98% of first turns run code and **0% produce a candidate answer**, while in the control arm a candidate answer exists in 56% of trajectories through turn 1 and 80% through turn 2, tracking the effect curve. Splitting the threatening-versus-neutral contrast by whether an answer existed yet, holding turn index fixed, gives +0.34 ($p = 0.27$, 31 tasks) against **+1.97** ($p = 0.0003$, 38 tasks) at turn 1, and +0.40 ($p = 0.65$, 8 tasks) against **+1.66** ($p = 0.0002$, 46 tasks) at turn 2. Holding the moderator fixed and moving the interjection changes little; holding the turn index fixed and varying the moderator changes everything. The no-answer cells are small, and tasks with no answer by turn 1 are also the harder tasks, so difficulty is not fully separated from the moderator.

This is the second timing account this study has withdrawn. The first was selection: **a timing effect of +21% at turn 1 against +39% at turn 2 ($p = 0.004$)** came from a turn-2 injection only firing in trajectories that reach turn 2, the harder and longer tasks; recomputed on a control-defined population it is +20.0% against +29.1%, $p = 0.32$. That is why injection turn is crossed and the comparison population fixed from control (§2.2). Three other claims were withdrawn on re-analysis: a ceiling explanation, tested and dropped; a "productive window" at turns 10–14 that was a rate on a small denominator — "zero of 43 improvements occurred at turn 15 or later" was about improvements to a trajectory's *running best*, and under the looser definition of any turn beating the one before it, 10 of 95 occur at turn 15 or later; and an earlier claim that the praise stop was demonstrably not premature, which the corrected regrade does not support (§5.3).

## C.3 The regrade instrument was wrong, and this is what changed

The first version of §6 reported a clean null: no arm moved the progress measure. That was an artefact. The regrader read each turn's output with openpyxl's `data_only=True` and **never recalculated it**, while the live grader had always passed output workbooks through LibreOffice first. A formula written by openpyxl carries no cached value, so every turn that answered with a formula read as empty and scored 0.0 — indistinguishable

from a turn that wrote nothing, and roughly 56–62% of Luna's final gradable turns write formulas. The section's own validation paragraph would have caught it: on the archived data **89 of 126 passed control trajectories scored exactly 0.0**. The check was described and not run.

| | Broken | Corrected, all scheduled trajectories | Corrected, primary population (§6) |
|---|---|---|---|
| Passed trajectories scoring 1.0 | 29.4% | **94.4%** | 94.4% |
| Redundant-step effects (demand / praise the assistant / closing cue) | +0.96 / −0.76 / −1.01 | +0.75 / −0.61 / −0.86 — all signs and significance held | +0.92 / −0.72 / −1.04 |
| "Work remains" on final match (three runs) | +0.014 / **+0.061** / +0.019 | +0.026 / +0.034 / +0.025 | +0.020 / **+0.051** / +0.009 |
| Conclusion | extra turns buy nothing | extra turns buy 2.8 points, replicated | extra turns buy about 2 points, not distinguishable from zero |

The redundant-step finding survived the fix with every sign and significance level intact, about 20% smaller on all scheduled trajectories. The bottom row is the reversal. Under the defective regrade the continue-signal arm's three final-match measurements read one hit between two nulls, spanning 4.4×, and were presented as this study's most consequential instance of run-to-run instability. **Without the fix we would have concluded that extra persistence buys nothing and that our own measurements of it were unstable**, from one unrecalculated workbook; the first conclusion is now doubtful and the second was wrong. That contrast's accuracy row is real instability and survives the fix; its apparent corroboration by the progress measure was never independent, since accuracy and `final_match` are two functions of the same workbook. The progress measure now carries a regression test pinning the failure directly — a workbook answering with `=A1*2` scores 0.0 without recalculation and 1.0 with it — but its redundant-step companion is the exact one, and readers should weight the two accordingly.

**A second correction, found in audit.** The corrected figures above were still measured on every scheduled trajectory, while the primary turn-count outcome excludes trajectories on which the interjection never fired and, in the register run, injection turn 0. §6 earlier divided a progress effect from the first population by a turn effect from the second, which gave "55–65% of each turn effect" and a pooled "work remains" gain of +2.8 points (95% CI [+0.5, +5.2], $p = 0.022$). On the population the turn-count outcome uses, redundant steps are **56–93%** of each clear turn effect and the pooled gain is **+2.2 points (95% CI −0.2 to +4.8, $p = 0.09$)**: the redundancy conclusion strengthens, and the progress gain no longer excludes zero. The audit also found that the validation paragraph's claim that the one anomalous task "contributes a zero difference to every arm" was false; §6.3 now reports the contrasts with and without it.

## C.4 Redundant steps, per arm

Contrasts against each run's own neutral control, paired and clustered by task (§2.3), on the trajectories the turn-count outcome uses (§6.1):

| Arm | Run | Δ redundant steps | $p$ |
|---|---|---|---|
| Demand only | probe, ceiling 10 | **+0.92** | <0.0001 |
| Demand only | stage 1, ceiling 20 | **+1.57** | 0.0012 |
| Demand only | stage 1 GLM, ceiling 20 | +0.26 | 0.087 |
| "Work remains" | praise run, ceiling 10 | **+1.53** | <0.0001 |
| "Work remains" | stage 1, ceiling 20 | **+4.66** | <0.0001 |
| "Work remains" | ceiling-20 run | **+5.20** | <0.0001 |
| Praise the assistant | praise run, ceiling 10 | **−0.72** | <0.0001 |
| Praise the work | praise run, ceiling 10 | **−0.61** | 0.0004 |

| Arm | Run | Δ redundant steps | $p$ |
|---|---|---|---|
| Closing cue | praise run, ceiling 10 | **−1.04** | <0.0001 |
| Insult only | probe, ceiling 10 | −0.06 | 0.70 |
| L7 threatening | seven-level, ceiling 10 | **+1.11** | <0.0001 |
| L3 polite | seven-level, ceiling 10 | **+1.47** | <0.0001 |
| L5 rude | seven-level, ceiling 10 | +0.05 | 0.77 |